

M. J. Biopharm Private Limited
Plot No. 18, International Biotech Park, Hinjawadi, Phase II, Pune 411057.

**Format for Attachment**

Title: Outcome of Recommendation action -II of deviation occurred during calibration of TOC Analyzer (Instrument ID: I/QC/018)

Attachment No: XVI

Ref. document No: DEV/QC/25/010

Outcome of Recommended Action II:

As per recommended action, hypothesis activity planned on same day i.e. 21/03/2026 and work allotted to analyst Emp. Id. 952 as earlier involved analyst Emp. Id. 615 was left for the day.

As per plan analyst Emp. Id. 952 verified all the pre requisites before initiation of preparation for System suitability test as per SOP, 'Title: Operation, Calibration and Maintenance of Shimadzu TOC Analyzer Model No.: TOC-L CPH With Lab Solution Software, SOP No.: SOP/DP/QC/045.

After ensuring the availability of all the prerequisites, Analyst taken the new vial of 1,4 -Benzoquinone USPRS, Lot No: R15690 (In house Vial No.: RS/002/23-003) and initiated the preparation of 1,4 - Benzoquinone standard, under the supervision of QC reviewer Emp.ID 616 and the details are as below;

Preparation of Benzoquinone stock standard solution (50000 ppb):

Weighed 7.508 mg of 1,4 -Benzoquinone & transferred in 100 ml volumetric flask, then added 50 ml of purified water in above 100 ml volumetric flask, sonicated for 30 seconds to dissolve completely and then make up the final volume to 100 ml.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Taken 1 ml of above stock standard (50000 ppb) by pipetting with 1 ml glass bulb pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with ultra-pure water.

Further, filled the freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) in 40 ml cleaned and dried TOC vial and kept vial in TOC autosampler tray.

As per the recommended action, injected the freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) as per below sequence in the TOC analyzer to evaluate the potential error with respect to uneven mixing or improper preparation of the standard solution and the activity was performed by following all applicable procedural requirements as per SOP/DP/QC/045 by selecting the latest calibration curve 'CAL_21032026.2026_03_21_11_54_09.cald'

Sr. No.	Name of Sample	No. of Sample to be injected
01	1,4 -Benzoquinone	01

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After completion of above run, verified the obtained results and noted that SST results obtained from freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) i.e. 79.76 % does not meet the acceptance (Limit: 85 % to 115 %), however obtained results are comparable with the SST results i.e. 76.26 % (Mean Area - 33.76, SD 0.3251 and CV 0.96 %) which was obtained from 'Re-Diluted Vial' during the earlier hypothesis testing.

Name of Vials	Hypothesis Testing - System Suitability Test Results			
	Recovery of 1,4-Benzoquinone	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Freshly prepared 1,4 - Benzoquinone	79.76 %	35.06	0.1652	0.47 %
(SST Limit: Between 85-115 %)				

Mean area of freshly prepared 1,4 – Benzoquinone (35.06) and mean area obtained from 'Re-Diluted Vial' (33.76) found at lower side as compare with the mean area of 1,4 – Benzoquinone obtained during initial analysis (48.19), hence verified the last calibration report to check the mean area of 1,4 – Benzoquinone and noted that, % recovery and mean area of 1,4 – Benzoquinone obtained during last calibration which was performed on 12/03/2026 was 98.44 % and 36.45 respectively and it is comparable with the mean area of freshly prepared 1,4 – Benzoquinone (35.06).

Since, mean area of 1,4 – Benzoquinone obtained during last calibration and current calibration from the freshly prepared 1,4 – Benzoquinone are comparable, however there was huge variation in the % recovery, as the % recovery obtained during last calibration was 98.44 % and in the current calibration it is 79.76 %.

Calculation of the % recovery dependent on the mean area of water (0 ppb) and Sucrose standard (500 ppb) obtained during the 2-Point calibration, hence verified the details of 2-Point calibration from the current calibration and last calibration, and details are as below;

Current Calibration: (Date: 21/03/2026)

Name of Curve: CAL_21032026.2026_03_21_11_54_09.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	5.432	0.1581	2.91 %
Two Point Calibration (500 ppb)	42.58	0.4564	1.07 %

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Last Calibration: (Date: 12/03/2026)

Name of Curve: CAL_12032026_01.2026_03_12_15_48_20.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	3.153	0.2727	8.65 %
Two Point Calibration (500 ppb)	36.98	0.2905	0.79 %

Based on the data evaluation of 2-Point calibration from the current calibration and last calibration, it was observed that, during the last calibration, mean area of Sucrose (36.98) (Refer Attachment-XIV) and 1,4 – Benzoquinone (36.45) (Refer Attachment-XV) found comparable with each other, therefore % recovery obtained during last calibration was 98.44 %, however there is variation in the mean area of Sucrose (42.58) and freshly prepared 1,4 – Benzoquinone (35.06) obtained during current calibration, which led to % recovery obtained during current calibration was 79.76 %.

Based on the evaluation of data and obtained results, it is observed that mean area of 1,4 – Benzoquinone (36.45) obtained during last calibration and mean area of freshly prepared 1,4 – Benzoquinone (35.06) obtained during latest calibration, mean area obtained from 'Re-Diluted Vial' (33.76) are found comparable with each other, however mean area obtained during initial run of current calibration (48.19), Same vial (48.91), Refilled Vial (44.27) are found varies from the mean area of freshly prepared 1,4 – Benzoquinone and area obtained during latest calibration.

Based on the outcome of investigation and obtained results, assignable cause for occurred non conformance could not be identified, however most probable cause for occurrence variation in the mean area of sucrose and 1,4 – Benzoquinone could be uneven mixing or improper preparation of the standard solution, therefore latest generated calibration curve could not be used for the system suitability test.

Recommended action III:

Based on the most probable cause, it is recommended to repeat the 2-Point calibration and system suitability test by using freshly prepared Sucrose and 1,4 – Benzoquinone standard solution as per below procedure under the supervision of QC reviewer.

Note: Newly generated calibration curve shall be used for the system suitability test.

Sucrose stock solution (50000 ppb):

Weigh accurately 11.88 mg of sucrose and transfer into the 100 mL volumetric flask which containing purified water, shake well and sonicate for 30 seconds to dissolve completely and dilute to 100 mL with purified water.

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Sucrose standard solution (500 ppb):

Take 1 ml of above stock standard (50000 ppb) by pipetting with Micropipette / Glass pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with Purified water.

Preparation of Benzoquinone stock standard solution (50000 ppb):

Weigh 7.5 mg of 1,4 -Benzoquinone & transfer in 100 ml volumetric flask, then add 50 ml of purified water in above 100 ml volumetric flask, sonicate for 30 seconds to dissolve completely and then make up the final volume to 100 ml.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Take 1 ml of above stock standard (50000 ppb) by pipetting with Micropipette/ Glass pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with TOC water.

Inject the freshly prepared 1,4 -Benzoquinone (500 ppb) solution as per below sequence.

Sr. No.	Name of Sample	No. of Sample to be injected
01	Sucrose (0 ppb and 500 ppb)	01
02	1,4 -Benzoquinone	01

Prepared By (User) (Sign & Date)	Reviewed By (User) (Sign & Date)	Reviewed By (QA) (Sign & Date)
 21/03/26	 21/03/26	 21/03/26